

All of Statistics: A Concise Course in Statistical Inference
Solutions by David A. Lee

All errors, typographical and substantive, and other offenses, are entirely my own.

Chapter 1: Probability

Import Packages

Please see the associated GitHub repo for all code and comments to simulations.
[LINK HERE]

Question: 1.10.1

Fill in the details of the proof of Theorem 1.8. Also, prove the monotone decreasing case.

Theorem (Wasserman 1.8) (Continuity of Probabilities)

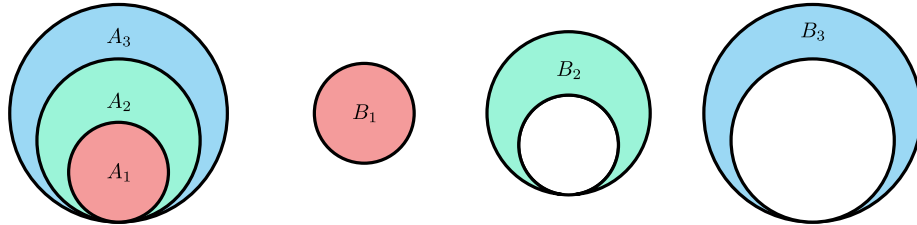
If $A_n \rightarrow A$ then

$$\mathbb{P}(A_n) \rightarrow \mathbb{P}(A)$$

as $n \rightarrow \infty$.

PROOF. Monotone Increasing Case. The intuition we exploit here is that monotonicity alone does not allow us to employ Kolmogorov Axiom 3. Therefore, we need to rewrite the monotone A 's in such a manner that the union of rewritten sets is disjoint.

Let A_n be monotone increasing. By definition, $A_1 \subset A_2 \subset \dots$. Let $A = \lim_{n \rightarrow \infty} A_n = \bigcup_{i=1}^{\infty} A_i$ and $B_1 = A_1$, $B_2 = \{\omega \in \Omega: \omega \in A_2, \omega \notin A_1\}$, and so on. Take any B_i, B_j , with $i < j$. Our intuition here is that each B_i consists of the "new" ω that is "added" to the previous A_{i-1} to create the new A_i .



Now, let $\omega \in B_i$. We must have $\omega \notin B_j$, since $\omega \notin B_i$ is a condition for membership in B_j . Thus $B_i \cap B_j = \emptyset$, and each of the B 's are pairwise disjoint. Since $A_1 \subset \dots \subset A_n$, we have

$$A_n = \bigcup_{i=1}^n A_i$$

Now consider $\bigcup_{i=1}^n A_i$. Let $\omega \in A_n$. Then $\omega \in B_n$ implies $\omega \in \bigcup_{i=1}^n B_n$. Conversely, let $\omega \in \bigcup_{i=1}^n B_n$. Then $\omega \in B_i$ for some i , meaning we must have $\omega \in A_i$; with $A_i \subset A_n$, we must have $\omega \in A_n$. Ergo, $A_n = \bigcup_{i=1}^n B_n$.

We now have established

$$A_n = \bigcup_{i=1}^n A_n = \bigcup_{i=1}^n B_n$$

and can conclude

$$\lim_{n \rightarrow \infty} A_n = \lim_{n \rightarrow \infty} \bigcup_{i=1}^n A_n = \lim_{n \rightarrow \infty} \bigcup_{i=1}^n B_n = A$$

By Kolmogorov Axiom 3 and the disjointness of the B 's, it follows that

$$P(A_n) = P\left(\bigcup_{i=1}^n B_i\right) = \sum_{i=1}^n P(B_i)$$

leaving the final step of taking the limits

$$\lim_{n \rightarrow \infty} P(A_n) = \lim_{n \rightarrow \infty} \sum_{i=1}^n P(B_i) = \sum_{i=1}^{\infty} P(B_i) = P\left(\bigcup_{i=1}^{\infty} B_i\right) = P(A)$$

Monotone Decreasing Case. With A_n monotone decreasing so that $A_1 \supset A_2 \supset \dots$, let $A = \lim_{n \rightarrow \infty} A_n = \bigcap_{i=1}^{\infty} A_i$. By de Morgan's laws, we have

$$\left(\bigcap_{i=1}^{\infty} A_i\right)^c = \bigcup_{i=1}^{\infty} A_i^c$$

and when applied to the monotone increasing case,

$$\lim_{n \rightarrow \infty} P(A_n^c) = 1 - \lim_{n \rightarrow \infty} P(A_n) = P\left(\bigcup_{i=1}^{\infty} A_i^c\right) = P(A^c) = 1 - P(A)$$

we can conclude

$$1 - \lim_{n \rightarrow \infty} P(A_n) = 1 - P(A) \implies \lim_{n \rightarrow \infty} P(A_n) = P(A)$$

□

Question: 1.10.2

Prove the statements in equation 1.1.

Property (Wasserman Eq. 1.1)

$$\begin{aligned}
 \mathbb{P}(\emptyset) &= 0 \\
 A \subset B &\implies \mathbb{P}(A) \leq \mathbb{P}(B) \\
 0 &\leq \mathbb{P}(A) \leq 1 \\
 \mathbb{P}(A^c) &= 1 - \mathbb{P}(A) \\
 A \cap B = \emptyset &\implies \mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)
 \end{aligned}$$

PROOF. Let A, B be events.

$$\mathbb{P}(\emptyset) = 0$$

We know that $\Omega^c = \emptyset$. Then $\Omega^c \cap \Omega = \emptyset$ and $\Omega^c \cup \Omega = \Omega$. Combining Kolmogorov Axioms 2 and 3, conclude $\mathbb{P}(\Omega) = \mathbb{P}(\Omega^c) + \mathbb{P}(\Omega) = \mathbb{P}(\emptyset) + \mathbb{P}(\Omega) = 1$. Another application of Axiom 2 yields $\mathbb{P}(\emptyset) = 1 - \mathbb{P}(\Omega) = 1 - 1 = 0$.

$$A \subset B \implies \mathbb{P}(A) \leq \mathbb{P}(B)$$

Consider A and $B - A$. We have $A \cap B - A = \emptyset$, as $\omega \in A$ implies $\omega \notin B - A$ and $\omega \in B - A$ implies $\omega \notin A$.

Now we prove $B = A \cup B - A$. If $\omega \in A \cup B - A$ either $\omega \in A$ or $\omega \in B - A$. If $\omega \in A$, then since $A \subset B$, we have $\omega \in B$. If $\omega \in B - A$, then $\omega \in B$. And if $\omega \in B$, then $\omega \in B - A$ implies $\omega \in A \cup B - A$. Thus $B = A \cup B - A$.

By Axiom 3,

$$\mathbb{P}(B) = \mathbb{P}(A \cup B - A) = \mathbb{P}(A) + \mathbb{P}(B - A)$$

since $\mathbb{P}(B - A) \geq 0$, this implies $\mathbb{P}(A) \leq \mathbb{P}(B)$.

$$0 \leq \mathbb{P}(A) \leq 1$$

By Axiom 1, $\mathbb{P}(A) \geq 0$. By Axioms 2 and 3, $\mathbb{P}(\Omega) = 1$. Then $\Omega = A \cup A^c$ and $A \cap A^c = \emptyset$, which in turn implies $\mathbb{P}(\Omega) = \mathbb{P}(A) + \mathbb{P}(A^c) = 1$. Since $\mathbb{P}(A^c) \geq 0$, we have $\mathbb{P}(A) \leq 1$. Thus $0 \leq \mathbb{P}(A) \leq 1$.

$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$$

Since $A^c \cap A = \emptyset$ and $A^c \cup A = \Omega$, by Axioms 2 and 3, we have $\mathbb{P}(\Omega) = 1 = \mathbb{P}(A) + \mathbb{P}(A^c)$. We conclude that $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.

$$A \cap B = \emptyset \implies \mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$$

By disjointness of A, B , by Axiom 3, $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$. $\square$

Question: 1.10.3

Let Ω be a sample space and let $A_1, A_2, \dots$, be events. Define $B_n = \bigcup_{i=n}^{\infty} A_i$ and $C_n = \bigcap_{i=n}^{\infty} A_i$.

- (a) Show that $B_1 \supset B_2 \supset \dots$ and that $C_1 \subset C_2 \subset \dots$.
- (b) Show that $\omega \in \bigcap_{n=1}^{\infty} B_n$ if and only if ω belongs to an infinite number of the events $A_1, A_2, \dots$.
- (c) Show that $\omega \in \bigcup_{n=1}^{\infty} C_n$ if and only if ω belongs to all the events $A_1, A_2, \dots$ except possibly a finite number of those events.

PROOF. (a) For the base case, we prove $B_1 \supset B_2$. If $\omega \in B_2 = \bigcup_{i=2}^{\infty} A_i$, then $\omega \in A_1 \cup \bigcup_{i=2}^{\infty} A_i = B_1$. Thus $B_1 \supset B_2$.

Now assume $B_1 \supset \dots \supset B_n$ is true. We want to prove that $B_1 \supset \dots \supset B_n \supset B_{n+1}$. It will suffice to show $B_n \supset B_{n+1}$. If $\omega \in B_{n+1} = \bigcup_{i=n+1}^{\infty} A_i$, then we have $\omega \in A_n \cup \bigcup_{i=n+1}^{\infty} A_i = B_n$, so $B_n \supset B_{n+1}$.

The proof for the C_n 's is analogous. For the base case, $C_1 \subset C_2$, if $\omega \in C_1 = \bigcap_{i=1}^{\infty} A_i$ then $\omega \in A_i$ for all i . Then we must have $\omega \in A_2, A_3, \dots$, and consequently, $\omega \in C_2 = \bigcap_{i=2}^{\infty} A_i$.

For the induction hypothesis, suppose that $C_1 \subset \dots \subset C_n$. We must show $C_n \subset C_{n+1}$. If $\omega \in C_n = \bigcap_{i=n}^{\infty} A_i$, then we have $\omega \in A_{n+1}, \dots$ implying $\omega \in C_{n+1} = \bigcap_{i=n+1}^{\infty} A_i$.

(b) ($\implies$) Let $\omega \in \bigcap_{n=1}^{\infty} B_n$. Suppose ω belongs only to a finite number of events A_i . Pick the A_i with the highest subscript i . Now consider $B_{i+1} = \bigcup_{j=i+1}^{\infty} A_j$. Our assumption of finitude implies $\omega \notin B_{i+1}$, a contradiction. We will come to this contradiction for any choice of A_i . Thus it must be that ω belongs to an infinite number of events A_i .

($\impliedby$) Now suppose ω belongs to an infinite number of events A_i . We must show that $\omega \in \bigcap_{n=1}^{\infty} B_n$.

For $n = 1$, we must have $\omega \in B_1$, as ω is in infinitely many A_i 's, and $B_1 = \bigcup_{i=1}^{\infty} A_i$.

Now suppose $\omega \in B_n = \bigcup_{i=n}^{\infty} A_i$. To demonstrate $\omega \in B_{n+1}$, we realize that we cannot have only $\omega \in A_n, \omega \notin A_{n+1}, \dots$ as this would contradict the infinitude of membership. Thus we must have membership in A_i 's with $i > n$. Ergo, $\omega \in B_{n+1}$.

By induction, $\omega \in B_i$ for all i . Thus $\omega \in \bigcap_{n=1}^{\infty} B_n$.

(c) ($\implies$) Let $\omega \in \bigcup_{n=1}^{\infty} C_n$. Then $\omega \in C_n = \bigcap_{i=n}^{\infty} A_i$ for at least one n . If $\omega \in C_1 = \bigcap_{i=1}^{\infty} A_i$, then ω belongs to all events $A_1, A_2, \dots$. Suppose otherwise that $\omega \in C_n$ for $n \neq 1$. Then ω lacks membership in at least one of the $A_1, \dots, A_{n+1}$; in other words, it lacks membership in a finite number of those events.

($\impliedby$) Let ω belong to all $A_1, A_2, \dots$ or otherwise lack membership in a finite number of A_i . In the first case, $\omega \in \bigcap_{i=1}^{\infty} A_i = C_1$ implies $\omega \in \bigcup_{n=1}^{\infty} C_n$. Otherwise, pick the A_i with $\omega \notin A_i$ and the highest i . Then we have $\omega \in \bigcap_{j=i+1}^{\infty} A_j = C_{i+1}$, and still $\omega \in \bigcup_{n=1}^{\infty} C_n$. $\square$

Question: 1.10.4

Let $\{A_i : i \in I\}$ be a collection of events where I is an arbitrary index set. Show that

$$\left(\bigcup_{i \in I} A_i\right)^c = \bigcap_{i \in I} A_i^c \quad \text{and} \quad \left(\bigcap_{i \in I} A_i\right)^c = \bigcup_{i \in I} A_i^c$$

Hint: First prove this for $I = \{1, \dots, n\}$.

PROOF. These are the famous De Morgan's Laws.

(1) Let $\omega \in \left(\bigcup_{i \in I} A_i\right)^c$. Suppose $I = \{1, \dots, n\}$. Then we have $\omega \notin \bigcup_{i \in I} A_i$, with the following chain of implications: $\omega \notin A_1, \dots, A_n$ to $\omega \in A_1^c, \dots, A_n^c$ to $\omega \in \bigcap_{i \in I} A_i^c$. Conversely, if $\omega \in \bigcap_{i \in I} A_i^c$, we must have $\omega \notin A_1, \dots, A_n$, implying $\omega \notin \bigcup_{i \in I} A_i$, allowing us to conclude $\omega \in \left(\bigcup_{i \in I} A_i\right)^c$.

That is the finite case. Suppose now that we prove the infinite case. In the base case, trivially $\omega \in A_1^c$ implies itself. If we assume it holds for $n-1$ A_i 's, apply the proof for the n A_i 's to prove the infinite case by induction.

(2) Let $\omega \in \left(\bigcap_{i \in I} A_i\right)^c$ for $i \in \{1, \dots, n\}$. Then $\omega \notin \bigcap_{i \in I} A_i$, meaning ω is bereft of membership for at least one A_i . It follows that $\omega \in A_i^c$, implying $\omega \in \bigcup_{i \in I} A_i^c$. Now suppose $\omega \in \bigcup_{i \in I} A_i^c$. Then $\omega \in A_i^c$ for at least one i . If so, we cannot have $\omega \in \bigcap_{i \in I} A_i$, so it must follow that $\omega \in \left(\bigcap_{i \in I} A_i\right)^c$. Thus $\left(\bigcap_{i \in I} A_i\right)^c = \bigcup_{i \in I} A_i^c$. $\square$

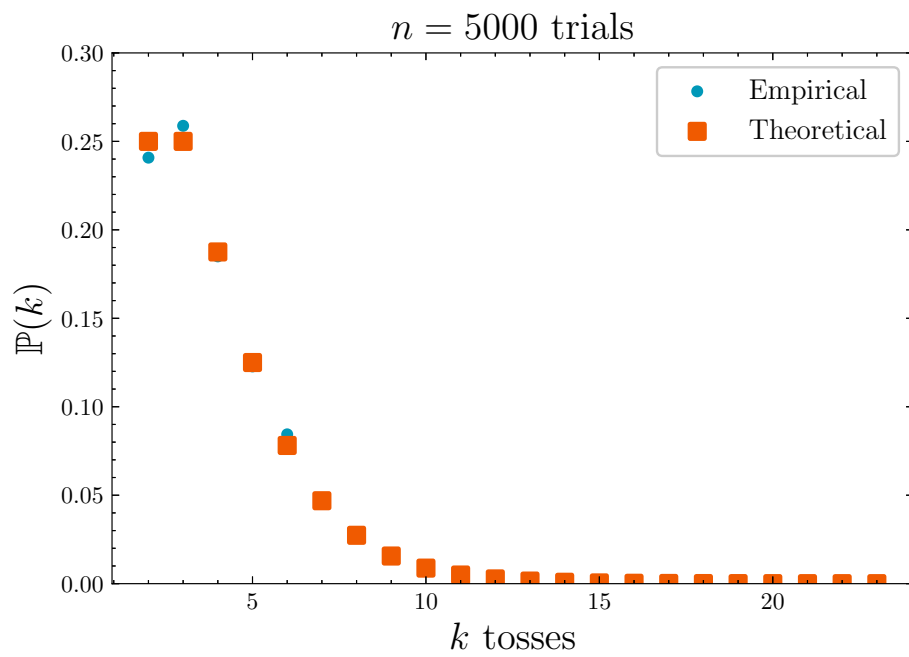
Question: 1.10.5

Suppose we toss a fair coin until we get exactly two heads. Describe the sample space S . What is the probability that exactly k tosses are required?

Descriptively, the sample space S is the set of events where the first head appears on any one of the $1, \dots, k-1$ -th toss. In general, the probability that exactly k tosses are needed is

$$\mathbb{P}(k \text{ tosses}) = (k-1)(1/2)^k$$

The reasoning is that since the second head is fixed on the k -th toss, there are $k-1$ tosses for the first head to be flipped. We simulate the coin toss experiment below:



When we simulate the experiment for a large number of trials, we can see that the empirical results are quite close to the expected theoretical outcomes.

Question: 1.10.6

Let $\Omega = \{0, 1, \dots\}$. Prove that there does not exist a uniform distribution on Ω (i.e., if $\mathbb{P}(A) = \mathbb{P}(B)$ whenever $|A| = |B|$, then $\mathbb{P}$ cannot satisfy the axioms of probability).

PROOF. Suppose that Ω has a uniform distribution, and $\mathbb{P}(A) = \mathbb{P}(B)$ whenever $|A| = |B|$. Consider the disjoint singletons $\{0\}, \{1\}, \dots$. By Axiom 3, we have

$$\mathbb{P}\left(\bigcup_{k=0}^{\infty} \{k\}\right) = \sum_{k=0}^{\infty} \mathbb{P}(\{k\})$$

However,

$$\bigcup_{k=0}^{\infty} \{k\} = \Omega$$

which from Axiom 2 implies

$$\mathbb{P}\left(\bigcup_{k=0}^{\infty} \{k\}\right) = \mathbb{P}(\Omega) = 1$$

But by our uniform distribution assumption, each constituent of the sum of singleton probabilities is zero (picking a particular integer out of infinite possibilities is effectively zero probability). Thus we must have

$$\sum_{k=0}^{\infty} \mathbb{P}(\{k\}) = 0$$

But $\mathbb{P}(\bigcup_{k=0}^{\infty} \{k\}) = 1$, a contradiction. $\square$

Question: 1.10.7

Let $A_1, A_2, \dots$ be events. Show that

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} \mathbb{P}(A_n).$$

Hint: Define $B_n = A_n - \bigcup_{i=1}^{n-1} A_i$. Then show that the B_n are disjoint and that $\bigcup_{n=1}^{\infty} A_n = \bigcup_{n=1}^{\infty} B_n$.

PROOF. Define $B_n = A_n - \bigcup_{i=1}^{n-1} A_i$. First we prove the B_n are disjoint. Take B_k, B_j for $k \neq j$. Without loss of generality, suppose $k < j$. Let $\omega \in B_k$. Then $\omega \in A_k - \bigcup_{i=1}^{k-1} A_i$; importantly, $\omega \in A_k$. Then we must have $\omega \in \bigcup_{i=1}^{j-1} A_i$, implying $\omega \notin A_j - \bigcup_{i=1}^{j-1} A_i = B_j$.

Conversely, if $\omega \in B_j = A_j - \bigcup_{i=1}^{j-1} A_i$, we must have $\omega \notin \bigcup_{i=1}^{j-1} A_i$. Then we have $\omega \notin A_k$, implying $\omega \notin A_k - \bigcup_{i=1}^{k-1} A_i$. Thus $B_i \cap B_j = \emptyset$ and B_i, B_j are disjoint.

Now, we establish $\bigcup_{n=1}^{\infty} A_n = \bigcup_{n=1}^{\infty} B_n$. Let $\omega \in \bigcup_{n=1}^{\infty} A_n$. Then $\omega \in A_k$ for some k (s). Choose the A_k with the lowest k . Then $\omega \in A_k - \bigcup_{i=1}^{k-1} A_i = B_k$ implies $\omega \in \bigcup_{n=1}^{\infty} B_n$.

Conversely, let $\omega \in \bigcup_{n=1}^{\infty} B_n$. Then $\omega \in B_k = A_k - \bigcup_{i=1}^{k-1} A_i$ for some k (s). Then $\omega \in A_k$ implies $\omega \in \bigcup_{n=1}^{\infty} A_n$. Thus $\bigcup_{n=1}^{\infty} A_n = \bigcup_{n=1}^{\infty} B_n$.

By Axiom 3 and the disjointness of B_n , we have

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} B_n\right) = \sum_{n=1}^{\infty} \mathbb{P}(B_n)$$

Since $B_n = A_n - \bigcup_{i=1}^{n-1} A_i \subset A_n$, we have $\mathbb{P}(B_n) \leq \mathbb{P}(A_n)$ for all n . Then $\sum_{n=1}^{\infty} \mathbb{P}(B_n) \leq \sum_{n=1}^{\infty} \mathbb{P}(A_n)$. At last we may conclude

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} \mathbb{P}(A_n)$$

□

Question: 1.10.8

Suppose that $\mathbb{P}(A_i) = 1$ for each i . Prove that

$$\mathbb{P}\left(\bigcap_{i=1}^{\infty} A_i\right) = 1$$

PROOF. By premise, $\mathbb{P}(A_i = 1)$ implies $\mathbb{P}(A_i^c) = 1 - \mathbb{P}(A_i) = 0$. Now, since $A_1^c, A_2^c, \dots$ all have probability zero, it must be the case that $\mathbb{P}(\bigcup_{i=1}^{\infty} A_i^c) = 0$. By De Morgan's laws,

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i^c\right) = \mathbb{P}\left(\left(\bigcap_{i=1}^{\infty} A_i\right)^c\right) = 0$$

Allowing us to conclude

$$\mathbb{P}\left(\bigcap_{i=1}^{\infty} A_i\right) = 1 - \mathbb{P}\left(\left(\bigcap_{i=1}^{\infty} A_i\right)^c\right) = 1$$

□

Question: 1.10.9

For fixed B such that $\mathbb{P}(B) > 0$, show that $\mathbb{P}(\cdot | B)$ satisfies the axioms of probability.

PROOF. (i) For independent $\cdot, B$, we have $\mathbb{P}(\cdot | B) = \mathbb{P}(\cdot)\mathbb{P}(B)$, and if $\mathbb{P}(\cdot) \geq 0$ then $\mathbb{P}(\cdot | B) = \mathbb{P}(\cdot) \geq 0$. Otherwise, $\mathbb{P}(\cdot | B), \mathbb{P}(B) \geq 0$ implies $\mathbb{P}(\cdot | B) \geq 0$.

(ii) If $\cdot = \Omega$, then

$$\mathbb{P}(\Omega \mid B) = \frac{\mathbb{P}(\Omega B)}{\mathbb{P}(B)} = \frac{\mathbb{P}(\Omega)\mathbb{P}(B)}{\mathbb{P}(B)} = \mathbb{P}(\Omega) = 1$$

(iii) If $A_i \cap A_j = \emptyset$, then $(A_i \cap B) \cap (A_j \cap B) = \emptyset$. Then we have

$$\begin{aligned} \mathbb{P}(A_i \cup A_j \mid B) &= \frac{\mathbb{P}((A_i \cup A_j) \cap B)}{\mathbb{P}(B)} \\ &= \frac{\mathbb{P}((A_i \cap B) \cup (A_j \cap B))}{\mathbb{P}(B)} \\ &= \frac{\mathbb{P}(A_i \cap B) + \mathbb{P}(A_j \cap B)}{\mathbb{P}(B)} \\ &= \mathbb{P}(A_i \mid B) + \mathbb{P}(A_j \mid B) \end{aligned}$$

To generalize for infinitely many disjoint events A_i , observe that

$$\left(\bigcup_{i=1}^{\infty} A_i \right) \cap B = \bigcup_{i=1}^{\infty} (A_i \cap B)$$

Applying the logic for the case with two disjoint events, we can immediately conclude

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i \mid B\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i \mid B)$$

□

Question: 1.10.10

You have probably heard it before. Now you can solve it rigorously. It is called the "Monty Hall Problem." A prize is placed at random behind one of three doors. You pick a door. To be concrete, let's suppose you always pick door 1. Now Monty Hall chooses one of the other two doors, opens it and shows you that it is empty. He then gives you the opportunity to keep your door or switch to the other unopened door. Should you stay or switch? Intuition suggests it doesn't matter. The correct answer is you should switch. Prove it. It will help to specify the sample space and the relevant events carefully. Thus write $\Omega = \{(\omega_1, \omega_2) : \omega_i \in \{1, 2, 3\}\}$ where ω_1 is where the prize is and ω_2 is the door Monty opens.

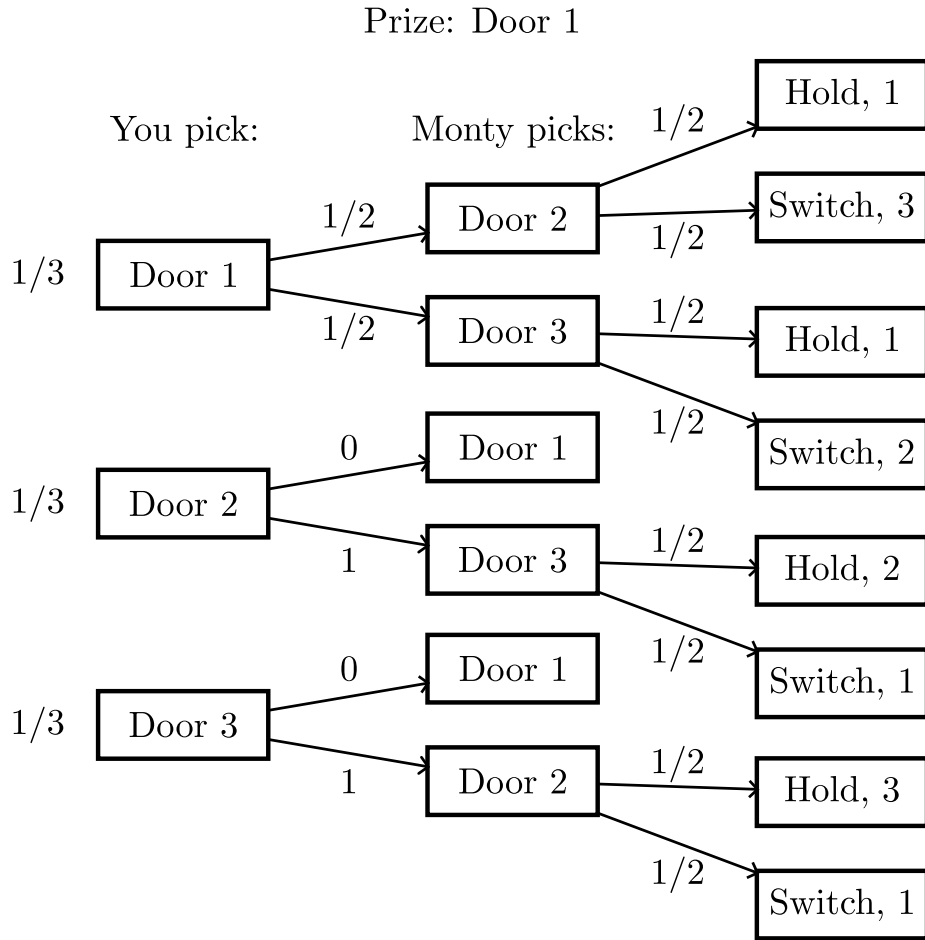
There are some insights we should be aware of. Firstly, there is equal chance of the prize appearing behind any one of the three doors, i.e., $\mathbb{P}(\text{Prize, Door } i) = 1/3$. Secondly, the choice of door for Monty Hall *depends on our choice of door*. Our pick of door also follows equal probability for each door, or $1/3$.

Using Bayes' Theorem, without loss of generality, suppose you pick door 1, the prize lies behind door 2, and Monty Hall picks door 3. We wish to find the probability that the prize is behind door 2, *given that Monty Hall opens door 3*.

$$\begin{aligned}\mathbb{P}(\text{Prize, Door 2} \mid \text{Monty, Door 3}) &= \frac{\mathbb{P}(\text{M, D 3} \mid \text{P, D 2})\mathbb{P}(\text{P, D 2})}{\sum \mathbb{P}(\text{M, D 3} \mid \text{P, D } i)\mathbb{P}(\text{P, D } i)} \\ &= \frac{1 \cdot 1/3}{1 \cdot 1/3 + 1/2 \cdot 1/3 + 0 \cdot 1/3} \\ &= \boxed{2/3}\end{aligned}$$

Here, it is advantageous to switch doors, as the prior probability ($1/3$) changes to the posterior probability ($2/3$) once Monty Hall diverges additional information that his door is empty.

It is useful to illustrate the sample space by way of a probability tree:



Assuming without loss of generality that the prize lies behind door 1, we can calculate the conditional probabilities of winning given that we either hold or switch.

Let Y be your choice of door and M be Monty Hall's. Moreover, let α be

$$\begin{aligned}
 \alpha &= \mathbb{P}(Y = 1)\mathbb{P}(M = 2 \mid Y = 1)\mathbb{P}(\text{Hold } 1 \mid M = 2) \\
 &\quad + \mathbb{P}(Y = 1)\mathbb{P}(M = 3 \mid Y = 1)\mathbb{P}(\text{Hold } 1 \mid M = 3) \\
 &\quad + \mathbb{P}(Y = 2)\mathbb{P}(M = 3 \mid Y = 2)\mathbb{P}(\text{Hold } 2 \mid M = 3) \\
 &\quad + \mathbb{P}(Y = 3)\mathbb{P}(M = 2 \mid Y = 3)\mathbb{P}(\text{Hold } 3 \mid M = 2) \\
 &= 1/3 \cdot 1/2 \cdot 1/2 + 1/3 \cdot 1/2 \cdot 1/2 + 1/3 \cdot 1/2 + 1/3 \cdot 1/2 \\
 &= 1/2
 \end{aligned}$$

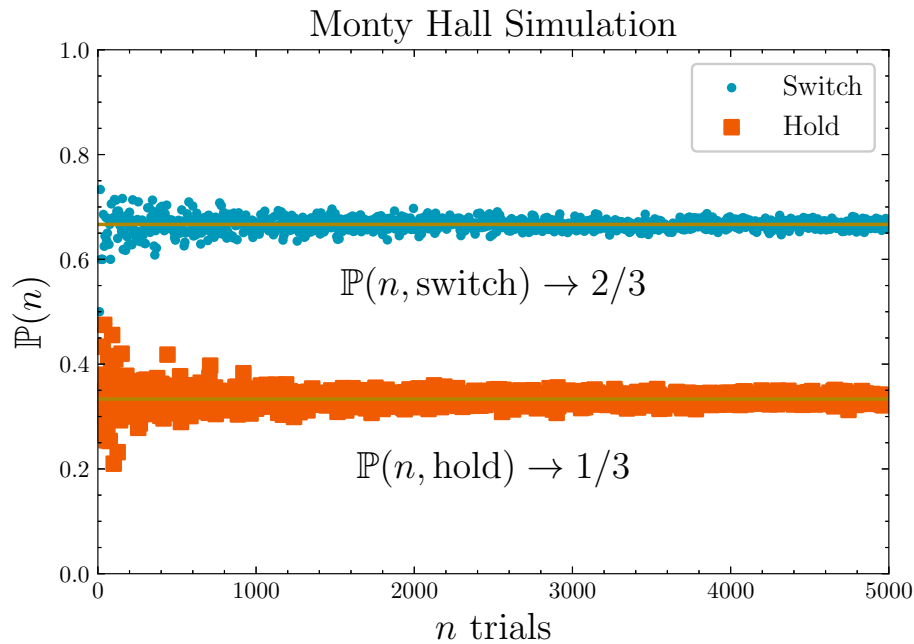
We calculate

$$\begin{aligned}
 \mathbb{P}(\text{Win} \mid \text{Hold}) &= \frac{\mathbb{P}(Y=1)\mathbb{P}(M=2 \mid Y=1)\mathbb{P}(\text{Hold } 1 \mid M=2)}{\alpha} \\
 &\quad + \frac{\mathbb{P}(Y=1)\mathbb{P}(M=3 \mid Y=1)\mathbb{P}(\text{Hold } 1 \mid M=3)}{\alpha} \\
 &= \frac{1/3 \cdot 1/2 \cdot 1/2 + 1/3 \cdot 1/2 \cdot 1/2}{1/2} \\
 &= \boxed{1/3}
 \end{aligned}$$

An analogous calculation for the switching conditional probability gives us

$$\mathbb{P}(\text{Win} \mid \text{Switch}) = \boxed{2/3}$$

Lastly, there is a famous anecdote of Erdős refusing to accept the result of the Monty Hall problem until examining a simulation of the game. You may find yourself burdened by a similar sentiment. Indeed, the experimental result aligns with our derivations above:



Question: 1.10.11

Suppose that A and B are independent events. Show that A^c and B^c are independent events.

PROOF. By premise, $\mathbb{P}(AB) = \mathbb{P}(A)\mathbb{P}(B)$. We have $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$ and $\mathbb{P}(B^c) = 1 - \mathbb{P}(B)$. To find $\mathbb{P}(A^c B^c)$, we observe these disjoint events:

$$\begin{aligned}
\mathbb{P}(A - B) &= \mathbb{P}(A) - \mathbb{P}(AB) \\
\mathbb{P}(B - A) &= \mathbb{P}(B) - \mathbb{P}(AB) \\
&\mathbb{P}(AB) \\
&\mathbb{P}(A^c B^c)
\end{aligned}$$

Thus we have

$$1 = \mathbb{P}(A - B) + \mathbb{P}(B - A) + \mathbb{P}(AB) + \mathbb{P}(A^c B^c)$$

which implies

$$\begin{aligned}
\mathbb{P}(A^c B^c) &= 1 - \mathbb{P}(A - B) - \mathbb{P}(B - A) - \mathbb{P}(AB) \\
&= 1 - (\mathbb{P}(A) - \mathbb{P}(AB)) - (\mathbb{P}(B) - \mathbb{P}(AB)) - \mathbb{P}(AB) \\
&= 1 - \mathbb{P}(A) - \mathbb{P}(B) + \mathbb{P}(AB)
\end{aligned}$$

Now calculate

$$\begin{aligned}
\mathbb{P}(A^c) \mathbb{P}(B^c) &= (1 - \mathbb{P}(A))(1 - \mathbb{P}(B)) \\
&= 1 - \mathbb{P}(A) - \mathbb{P}(B) + \mathbb{P}(A) \mathbb{P}(B) \\
&= 1 - \mathbb{P}(A) - \mathbb{P}(B) + \mathbb{P}(AB)
\end{aligned}$$

Thus $\mathbb{P}(A^c B^c) = \mathbb{P}(A^c) \mathbb{P}(B^c)$, and A^c and B^c are independent. $\square$

Question: 1.10.12

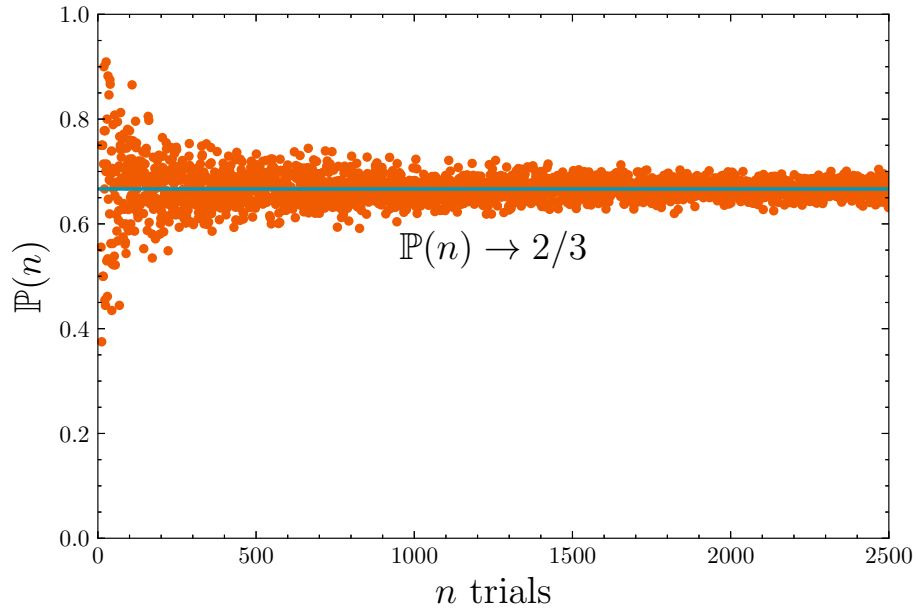
There are three cards. The first is green on both sides, the second is red on both sides and the third is green on one side and red on the other. We choose a card at random and we see one side (also chosen at random). If the side we see is green, what is the probability that the other side is also green? Many people intuitively answer $1/2$. Show that the correct answer is $2/3$.

PROOF. Let GG, RR, GR correspond to the first, second, and third cards. Using Bayes' Theorem: we calculate the probability of having drawn a GG card given that we see G , or GR given that we see G . Calculate

$$\begin{aligned}
\mathbb{P}(GG | G) &= \frac{\mathbb{P}(G | GG) \mathbb{P}(GG)}{\mathbb{P}(G | GG) \mathbb{P}(GG) + \mathbb{P}(G | RR) \mathbb{P}(RR) + \mathbb{P}(G | GR) \mathbb{P}(GR)} \\
&= \frac{1 \cdot 1/3}{1 \cdot 1/3 + 0 \cdot 1/3 + 1/2 \cdot 1/3} \\
&= \boxed{2/3} \\
\mathbb{P}(GR | G) &= \frac{\mathbb{P}(G | GR) \mathbb{P}(GR)}{\mathbb{P}(G | GR) \mathbb{P}(GR) + \mathbb{P}(G | RR) \mathbb{P}(RR) + \mathbb{P}(G | GG) \mathbb{P}(GG)} \\
&= \frac{1/2 \cdot 1/3}{1/2 \cdot 1/3 + 0 \cdot 1/3 + 1 \cdot 1/3} \\
&= \boxed{1/3}
\end{aligned}$$

$\square$

Simulating the card game, we find that the empirical outcome agrees with our theoretical result.



Question: 1.10.13

Suppose that a fair coin is tossed repeatedly until both a head and tail have appeared at least once.

- (a) Describe the sample space Ω .
- (b) What is the probability that three tosses will be required?

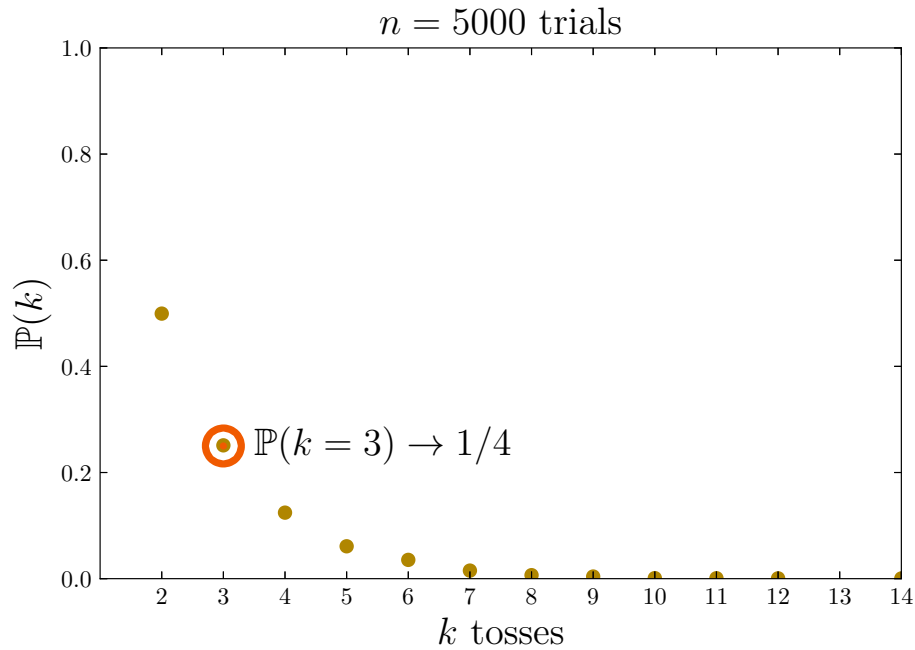
- (a) The sample space is given by

$$\Omega = \{HT, HHT, \dots, \underbrace{H \cdots H}_k T, \dots, TH, TTH, \dots, \underbrace{T \cdots T}_k H, \dots\}$$

- (b) The theoretical derivation of the probability is

$$\begin{aligned} \mathbb{P}(3 \text{ tosses}) &= \mathbb{P}(HHT) + \mathbb{P}(TTH) \\ &= (1/2)^3 + (1/2)^3 \\ &= \boxed{1/4} \end{aligned}$$

By simulation, we can observe the empirical probability of three tosses aligns with the theoretical result, as well as the probabilities for other numbers of tosses:

**Question: 1.10.14**

Show that if $\mathbb{P}(A) = 0$ or $\mathbb{P}(A) = 1$ then A is independent of every other event. Show that if A is independent of itself then $\mathbb{P}(A)$ is either 0 or 1.

PROOF. Let B be any event. Then if $\mathbb{P}(A) = 0$, we have $\mathbb{P}(AB) = 0$. Put differently, if A can never happen, then surely A and B could never happen. Now, it must also be true that $\mathbb{P}(A)\mathbb{P}(B) = 0$, so it follows that $\mathbb{P}(AB) = \mathbb{P}(A)\mathbb{P}(B) = 0$ and A is independent of all events.

Now suppose $\mathbb{P}(A) = 1$. Then $\mathbb{P}(AB) = \mathbb{P}(B)$. The reasoning is that since A will *always* happen, whether A and B happen together is contingent on B happening. Thus we can write $\mathbb{P}(AB) = \mathbb{P}(A)\mathbb{P}(B) = \mathbb{P}(B)$, again proving independence of A relative to all other events.

Lastly, suppose A is independent of itself. Then $\mathbb{P}(AA) = \mathbb{P}(A) = \mathbb{P}(A)\mathbb{P}(A)$, implying $\mathbb{P}(A) = 0$ or $\mathbb{P}(A) = 1$. $\square$

Question: 1.10.15

The probability that a child has blue eyes is $1/4$. Assume independence between children. Consider a family with 3 children.

- (a) If it is known that at least one child has blue eyes, what is the probability that at least two children have blue eyes?
- (b) If it is known that the youngest child has blue eyes, what is the probability that at least two children have blue eyes?

(a) The complement event to "at least one child has blue eyes" – call it A – is "no child has blue eyes." The probability that a child does not have blue eyes is $1 - p = 3/4$. Then we have

$$\mathbb{P}(A) = 1 - (3/4)^3 = 1 - 27/64 = 37/64$$

The probability that "at least two children have blue eyes" – call this one B – is the sum of the mutually exclusive probabilities of either two or three children having blue eyes. This is given by

$$\begin{aligned}\mathbb{P}(B) &= \mathbb{P}(2 \text{ children} + 3 \text{ children}) \\ &= \binom{3}{2} (1/4)^2 (3/4) + (1/4)^3 \\ &= 9/64 + 1/64 \\ &= 10/64\end{aligned}$$

Now we can calculate the probability of at least two children having blue eyes, given that at least one child has blue eyes.

$$\begin{aligned}\mathbb{P}(B | A) &= \frac{\mathbb{P}(BA)}{\mathbb{P}(A)} \\ &= (10/64)(64/37) \\ &= \boxed{10/37}\end{aligned}$$

(b) The probability that the youngest child has blue eyes (call it C) is $1/4$. Let D be the event that at least two children have blue eyes, with one of them being the youngest child. Then we have

$$\begin{aligned}\mathbb{P}(D) &= 2(1/4)^2 (3/4) + (1/4)^3 \\ &= 6/64 + 1/64 \\ &= 7/64\end{aligned}$$

Lastly, the probability that at least two children have blue eyes given that the youngest child is known to have blue eyes is

$$\begin{aligned}\mathbb{P}(D | C) &= \frac{\mathbb{P}(DC)}{\mathbb{P}(C)} \\ &= \frac{7/64}{1/4} \\ &= \boxed{7/16}\end{aligned}$$

Question: 1.10.16

Prove Lemma 1.14.

Lemma (Wasserman 1.14)

If A and B are independent events then $\mathbb{P}(A | B) = \mathbb{P}(A)$. Also, for any pair of events A and B ,

$$\mathbb{P}(AB) = \mathbb{P}(A | B)\mathbb{P}(B) = \mathbb{P}(B | A)\mathbb{P}(A)$$

PROOF.

$$\mathbb{P}(A | B) = \frac{\mathbb{P}(AB)}{\mathbb{P}(B)} = \frac{\mathbb{P}(A)\mathbb{P}(B)}{\mathbb{P}(B)} = \mathbb{P}(A)$$

$$\mathbb{P}(A | B) = \frac{\mathbb{P}(AB)}{\mathbb{P}(B)} \implies \mathbb{P}(A | B)\mathbb{P}(B) = \mathbb{P}(AB)$$

$$\mathbb{P}(B | A) = \frac{\mathbb{P}(AB)}{\mathbb{P}(A)} \implies \mathbb{P}(B | A)\mathbb{P}(A) = \mathbb{P}(AB)$$

□

Question: 1.10.17

Show that

$$\mathbb{P}(ABC) = \mathbb{P}(A | BC)\mathbb{P}(B | C)\mathbb{P}(C)$$

PROOF.

$$\mathbb{P}(ABC) = \mathbb{P}(A | BC)\mathbb{P}(BC) = \mathbb{P}(A | BC)\mathbb{P}(B | C)\mathbb{P}(C)$$

□

Question: 1.10.18

Suppose k events form a partition of the sample space Ω , i.e., they are disjoint and $\bigcup_{i=1}^k A_i = \Omega$. Assume that $\mathbb{P}(B) > 0$. Prove that if $\mathbb{P}(A_1 | B) < \mathbb{P}(A_1)$ then $\mathbb{P}(A_i | B) > \mathbb{P}(A_i)$ for some $i = 2, \dots, k$.

PROOF. By disjointness of events, we have

$$\sum_{i=1}^k \mathbb{P}(A_i) = 1 \quad \text{and} \quad \sum_{i=1}^k \mathbb{P}(A_i | B) = 1$$

with the second equality following from 1.10.9. Let $\mathbb{P}(A_1 | B) < \mathbb{P}(A_1)$. It cannot be the case that we have $\mathbb{P}(A_i | B) < \mathbb{P}(A_i)$ for the remaining $i = 2, \dots, k$, for this would violate additivity of the conditional probabilities to unity:

$$\sum_{i=2}^k \mathbb{P}(A_i | B) < \sum_{i=2}^k \mathbb{P}(A_i) = 1$$

So would the condition that $\mathbb{P}(A_i | B) = \mathbb{P}(A_i)$ for the remaining i . Thus at least one of the $\mathbb{P}(A_i | B)$ must be greater than A_i for the unity sum condition to be met. $\square$

Question: 1.10.19

Suppose that 30 percent of computer owners use a Macintosh, 50 percent use Windows, and 20 percent use Linux. Suppose that 65 percent of the Mac users have succumbed to a computer virus, 82 percent of the Windows users get the virus, and 50 percent of the Linux users get the virus. We select a person at random and learn that her system was infected with the virus. What is the probability that she is a Windows user?

By Bayes' Theorem:

$$\begin{aligned}\mathbb{P}(\text{Windows} | \text{Virus}) &= \frac{\mathbb{P}(V | W)\mathbb{P}(W)}{\mathbb{P}(V | W)\mathbb{P}(W) + \mathbb{P}(V | M)\mathbb{P}(M) + \mathbb{P}(V | L)\mathbb{P}(L)} \\ &= \frac{0.82 \cdot 0.5}{0.82 \cdot 0.5 + 0.65 \cdot 0.3 + 0.5 \cdot 0.2} \\ &= \boxed{0.582}\end{aligned}$$

Question: 1.10.20

A box contains 5 coins and each has a different probability of showing heads. Let $p_1, \dots, p_5$ denote the probability of heads on each coin. Suppose that

$$p_1 = 0, p_2 = 1/4, p_3 = 1/2, p_4 = 3/4 \text{ and } p_5 = 1.$$

Let H denote "heads is obtained" and let C_i denote the event that coin i is selected.

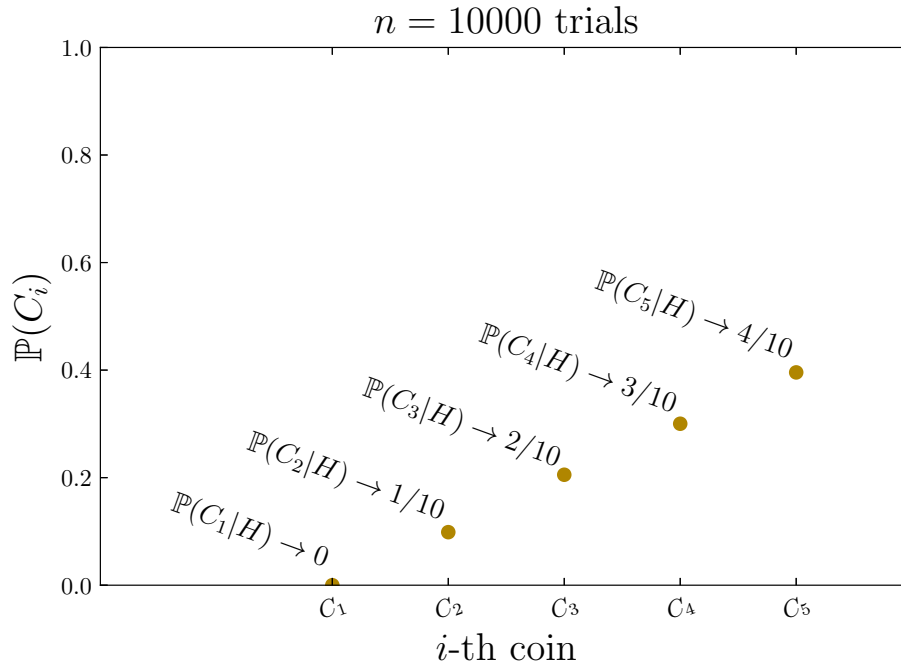
- Select a coin at random and toss it. Suppose a head is obtained. What is the posterior probability that coin i was selected ($i = 1, \dots, 5$)? In other words, find $\mathbb{P}(C_i | H)$ for $i = 1, \dots, 5$.
- Toss the coin again. What is the probability of another head? In other words find $\mathbb{P}(H_2 | H_1)$ where $H_j = \text{"heads on toss } j\text{"}$.
Now suppose that the experiment was carried out as follows: We select a coin at random and toss it until a head is obtained.
- Find $\mathbb{P}(C_i | B_4)$ where $B_4 = \text{"first head is obtained in toss 4."}$

Apply Bayes' Theorem for all parts.

- Here, $\sum \mathbb{P}(H | C_i)\mathbb{P}(C_i) = 1/5 \cdot \left(\sum p_i\right) = 1/5 \cdot 5/2 = 1/2$.

$$\begin{aligned}
\mathbb{P}(C_1 | H) &= \frac{\mathbb{P}(H | C_1)\mathbb{P}(C_1)}{\sum \mathbb{P}(H | C_i)\mathbb{P}(C_i)} = \boxed{0} \\
\mathbb{P}(C_2 | H) &= \frac{\mathbb{P}(H | C_2)\mathbb{P}(C_2)}{\sum \mathbb{P}(H | C_i)\mathbb{P}(C_i)} = \frac{1/4 \cdot 1/5}{1/2} = \boxed{1/10} \\
\mathbb{P}(C_3 | H) &= \frac{\mathbb{P}(H | C_3)\mathbb{P}(C_3)}{\sum \mathbb{P}(H | C_i)\mathbb{P}(C_i)} = \frac{1/2 \cdot 1/5}{1/2} = \boxed{2/10} \\
\mathbb{P}(C_4 | H) &= \frac{\mathbb{P}(H | C_4)\mathbb{P}(C_4)}{\sum \mathbb{P}(H | C_i)\mathbb{P}(C_i)} = \frac{3/4 \cdot 1/5}{1/2} = \boxed{3/10} \\
\mathbb{P}(C_5 | H) &= \frac{\mathbb{P}(H | C_5)\mathbb{P}(C_5)}{\sum \mathbb{P}(H | C_i)\mathbb{P}(C_i)} = \frac{1 \cdot 1/5}{1/2} = \boxed{4/10}
\end{aligned}$$

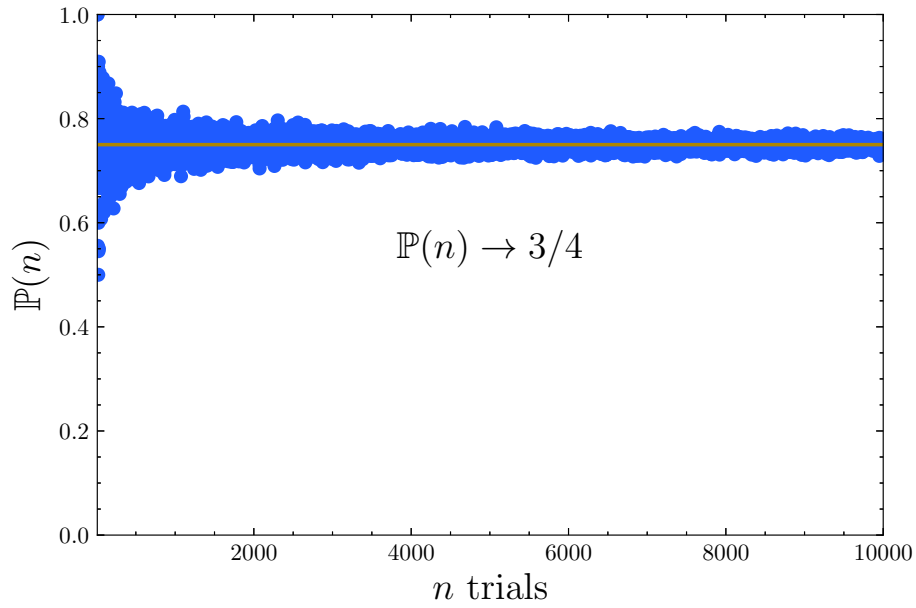
Empirical simulation:



(b) By the Law of Total Probability, $\mathbb{P}(H_1) = 1/5 \cdot \sum p_i = 1/2$. To calculate $\mathbb{P}(H_2 | H_1)$, write

$$\begin{aligned}
\mathbb{P}(H_2 | H_1) &= \frac{\mathbb{P}(H_2 H_1)}{\mathbb{P}(H_1)} \\
&= \frac{1/5 \cdot \sum p_i^2}{1/2} \\
&= \frac{3/8}{1/2} \\
&= \boxed{3/4}
\end{aligned}$$

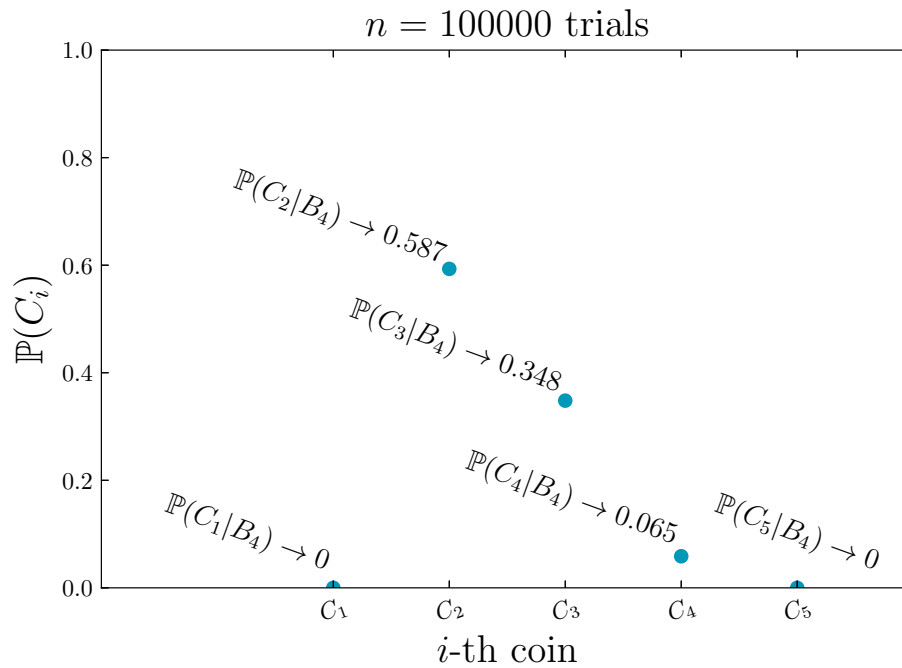
Empirical simulation:



(c) Here, $\sum \mathbb{P}(B_4 | C_i) \mathbb{P}(C_i) = 1/5 \cdot \sum (1 - p_i)^3 p_i = 0.0359375$.

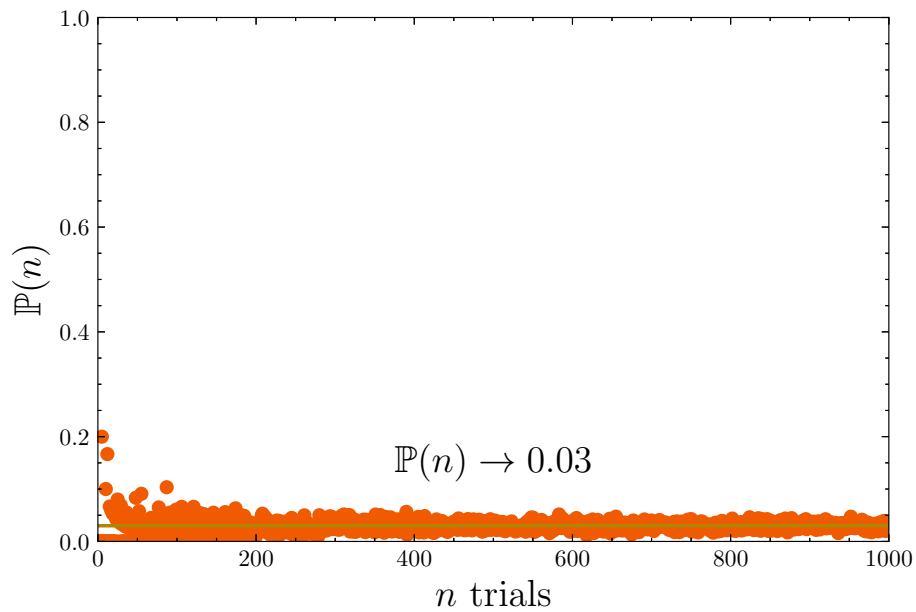
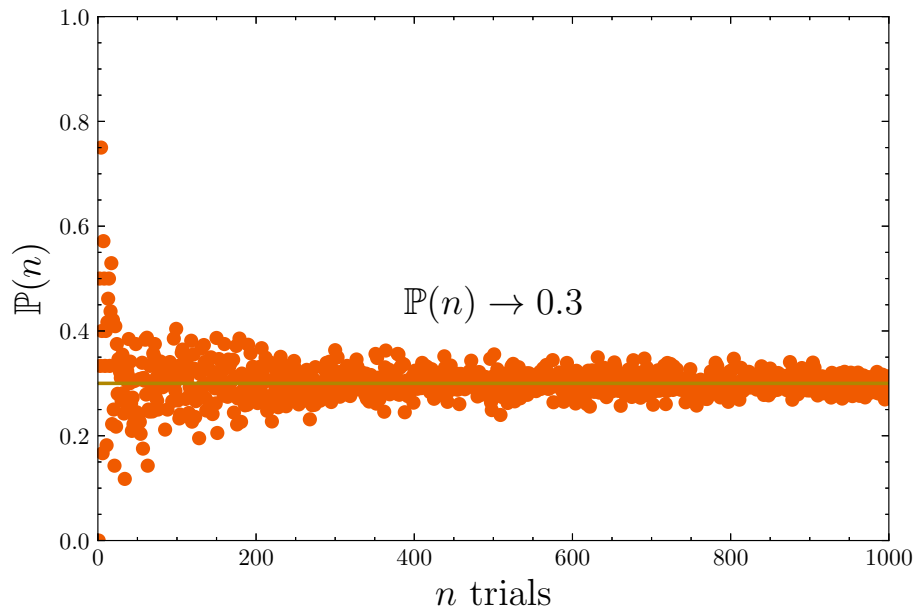
$$\begin{aligned}
 \mathbb{P}(C_1 | B_4) &= \frac{\mathbb{P}(B_4 | C_1) \mathbb{P}(C_1)}{\sum \mathbb{P}(B_4 | C_i) \mathbb{P}(C_i)} = \boxed{0} \\
 \mathbb{P}(C_2 | B_4) &= \frac{\mathbb{P}(B_4 | C_2) \mathbb{P}(C_2)}{\sum \mathbb{P}(B_4 | C_i) \mathbb{P}(C_i)} = \frac{(3/4)^3 (1/4) (1/5)}{0.0359375} = \boxed{0.587} \\
 \mathbb{P}(C_3 | B_4) &= \frac{\mathbb{P}(B_4 | C_3) \mathbb{P}(C_3)}{\sum \mathbb{P}(B_4 | C_i) \mathbb{P}(C_i)} = \frac{(1/2)^4 (1/5)}{0.0359375} = \boxed{0.348} \\
 \mathbb{P}(C_4 | B_4) &= \frac{\mathbb{P}(B_4 | C_4) \mathbb{P}(C_4)}{\sum \mathbb{P}(B_4 | C_i) \mathbb{P}(C_i)} = \frac{(1/4)^3 (3/4) (1/5)}{0.0359375} = \boxed{0.065} \\
 \mathbb{P}(C_5 | B_4) &= \frac{\mathbb{P}(B_4 | C_5) \mathbb{P}(C_5)}{\sum \mathbb{P}(B_4 | C_i) \mathbb{P}(C_i)} = \boxed{0}
 \end{aligned}$$

Empirical simulation:

**Question: 1.10.21**

(Computer Experiment.) Suppose a coin has probability p of falling heads up. If we flip the coin many times, we would expect the proportion of heads to be near p . We will make this formal later. Take $p = .3$ and $n = 1,000$ and simulate n coin flips. Plot the proportion of heads as a function of n . Repeat for $p = .03$.

As intuitively anticipated, for large n , the probability of heads tends towards 0.3 or 0.03.

**Question: 1.10.22**

(Computer Experiment.) Suppose we flip a coin n times and let p denote the probability of heads. Let X be the number of heads. We call X a binomial random variable, which is discussed in the next chapter. Intuition suggests that X will be close to np . To see if this is true, we can repeat this experiment many times and average the X values. Carry out a simulation and compare the average of the X 's to np . Try this for $p = .3$ and $n = 10, n = 100$, and $n = 1,000$.

We run 1,000 trials for 10, 100, and 1,000 flips. The average number of heads for each number of flips are 3.009, 30.13, and 299.746, compared to $np = 3, 30,$ and 300.

Question: 1.10.23

(Computer Experiment.) Here we will get some experience simulating conditional probabilities. Consider tossing a fair die. Let $A = \{2, 4, 6\}$ and $B = \{1, 2, 3, 4\}$. Then, $\mathbb{P}(A) = 1/2$, $\mathbb{P}(B) = 2/3$ and $\mathbb{P}(AB) = 1/3$. Since $\mathbb{P}(AB) = \mathbb{P}(A)\mathbb{P}(B)$, the events A and B are independent. Simulate draws from the sample space and verify that $\hat{\mathbb{P}}(AB) = \hat{\mathbb{P}}(A)\hat{\mathbb{P}}(B)$ where $\hat{\mathbb{P}}(A)$ is the proportion of times A occurred in the simulation and similarly for $\hat{\mathbb{P}}(AB)$ and $\hat{\mathbb{P}}(B)$. Now find two events A and B that are not independent. Compute $\hat{\mathbb{P}}(A)$, $\hat{\mathbb{P}}(B)$ and $\hat{\mathbb{P}}(AB)$. Compare the calculated values to their theoretical values. Report your results and interpret.

Now suppose we have $A = B = \{k\}$ where k is any integer $1, \dots, 6$. Then $\mathbb{P}(A) = \mathbb{P}(B) = \mathbb{P}(AB) = 1/6$, but $\mathbb{P}(A)\mathbb{P}(B) = 1/36$. Thus A and B are not independent. In general, we can conclude that any event cannot be independent of itself unless the probability of the event is either 0 or 1.